

Ivy Liu (刘瑗玮)

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🏠 [iiivliu.github.io](https://github.com/iiivliu)

🎓 EDUCATION

University of Wisconsin-Madison

Madison, Wisconsin, USA

M.A. in Mathematics

Sep. 2025 - Jun. 2027 (expected)

- GPA: 4.0/4.0
- Scholarship: Academic Year 2026-2027 VISP & Master's Scholarship (awarded to top 10% students)
- Selected coursework: Algebraic Geometry (schemes and sheaves), Abstract Algebra (categories, representation theory, commutative algebra, Galois theory), Combinatorics (algebraic graph theory).

Shanghai Jiao Tong University

Shanghai, China

B.S. in Mathematics and Applied Mathematics

Sep. 2022 - Jun. 2026

- Obtained A/A+ grades in all six algebra-related courses taken: *Advanced Algebra I, Advanced Algebra II, Elementary Number Theory, Abstract Algebra, Algebraic Geometry, Coding and Decoding*
- Thesis: *Theory of Toric Varieties and Its Application to the Face Enumeration of Simplicial Polytopes*
- Scholarship: Guo Chenchen Scholarship, 2023

🔍 RESEARCH EXPERIENCE

Entrywise Closure of Adjacency Algebras of Q -polynomial Graphs

July. 2026 - Present

Advised by Prof. Paul Terwilliger

University of Wisconsin-Madison

- Disproved a conjecture proposed by Paul Terwilliger in his paper *Distance-regular graphs, the subconstituent algebra, and the Q -polynomial property* that the adjacency algebra of every Q -polynomial graph is closed under entrywise product, by constructing an infinite family of counterexamples via Cartesian products, including one for every diameter $D \geq 3$.
- Formulated a structural refinement of the conjecture motivated by the counterexamples, and proved it for all Abelian Cayley graphs using character theory and additive combinatorics.
- Manuscript in preparation (sole author).

Bounded Ratio of Lorentzian Polynomials

Jan. 2026 - Present

Advised by Prof. Botong Wang

University of Wisconsin-Madison

- Read Brändén and Huh's paper *Lorentzian Polynomials* in a small directed reading group, and presented the proof that the space of Lorentzian Polynomials is closed.
- Research on the bounded ratio of Lorentzian Polynomials. Preprints (with Dijia Chen, Bowen Gan, Zemeng Wang, and Chengzhi Wu):
 - * Bounded Ratios of Lorentzian Polynomials I: The Ternary Theory and Optimal Bounding Constants.
 - * Bounded Ratios of Lorentzian Polynomials II: The Complete Quadratic Local-to-Global Classification.

PKU Algebraic Combinatorics Experience

Jul. 2024 - Aug. 2024

Deodhar Diagrams and Rational Dyck Paths

Advised by Prof. Long Guo and Prof. Yibo Gao

Peking University

- Worked on a problem of Galashin and Lam by establishing a natural bijection between rational Dyck paths and Deodhar diagrams for parameters $(n, kn \pm 1)$, and interpreting the area statistic for the $(n, n + 1)$ case.
- Enumerated Deodhar diagrams in the non-coprime case and introduced the associated Elbow and Pipe graphs, formulating related conjectures.

Seminar on Modular Forms

Jan. 2025 - Apr. 2025

- Studied Chapters 1-3.5, 6-7 of GTM 228 *A First Course in Modular Forms*, covering modular form, modular curve, the dimension formula of the space of the modular forms, Jacobian varieties and Abelian varieties.
- Gave a talk on the Hecke operators and their properties (Chapter 5) in the seminar, and the canonical basis of the space of the cusp forms.

Directed Study on Representations of Dynkin Quivers

May. 2024 - April. 2025

Advised by Prof. Pu Zhang

Shanghai Jiao Tong University

- Investigated theory of quivers and their representations, and Gabriel's theorem. Computed the minimal projective resolution of several classes of indecomposable modules in the category of finite-dimensional representations of Dynkin Quivers.

Seminar on Combinatorics Related to Learning Theory

Mar. 2024 - May. 2024

Advised by Prof. Yaokun Wu

- Studied part of the book *Sampling in Combinatorial and Geometric Set Systems*, covering ϵ -nets, VC-dimension and some related bounds.
- Acquired knowledge of some related topics, such as Karp's Algorithm for the maximum cycle mean problem.

SELECTED PRESENTATIONS

May. 15, 2026 - Theory of Toric Varieties and Its Application to the Face Enumeration of Simplicial Polytopes

Apr. 27, 2026 - On the Finite Dimensional Irreducible $\boxtimes^1$ -Modules

Mar. 13/20, 2026 - Closedness of the Space of Lorentzian Polynomials

Nov. 5, 2025 - On the Value Function Methods

Apr. 4, 2025 - Representations of Dynkin Quivers

Mar. 16, 2025 - The Slice Rank of a Direct Sum

Dec. 31, 2024 - On the MDS Conjecture

Nov. 16, 2024 - Deodhar Diagrams and Rational Dyck Paths

Apr. 22, 2023 - A More General Statement of Riesz Representation Theorem

¹ $\boxtimes$ refers to Tetrahedron algebra—not a typo.